

Question 1: Define social welfare. Critically explain the following criteria as a measure of social welfare: (i) Growth of GNP, (ii) Bentham's Criteria, (iii) Cardinalist approach.

Social welfare refers to the overall well-being of a society as a whole. Welfare economics, as defined by Koutsoyiannis, is concerned with the evaluation of alternative economic situations (states, configurations) from the point of view of the society's well-being. To illustrate, suppose total welfare in a country is W , but given factor endowments and technology, welfare could potentially be larger, say W^* . The tasks of welfare economics are: (a) to show that in the present state $W < W^*$, and (b) to suggest ways of raising W to W^* . The measurement of social welfare requires some ethical standard and interpersonal comparisons, both of which involve subjective value judgements. Objective comparisons of the deservingness or worthiness of different individuals are virtually impossible. Various criteria of social welfare have been suggested by economists at different times.

(i) Growth of GNP

Adam Smith implicitly accepted the growth of the wealth of a society — that is, the growth of the gross national product — as a welfare criterion. He believed that economic growth resulted in an increase in social welfare because growth increased employment and the goods available for consumption by the community. To Adam Smith, economic growth meant bringing W closer to W^* .

The growth criterion, however, implies an acceptance of the status quo of income distribution as 'ethical' or 'just'. This is a strong normative assumption that may not be defensible. Consider a society where economic growth is driven by a small elite while the majority of the population experiences stagnant or declining real incomes. The aggregate GNP may increase substantially, but if the distribution of the additional income is highly skewed, the majority may experience a deterioration in their well-being despite the growing GNP.

Furthermore, growth may lead to a reduction in social welfare depending on who benefits most from it. For example, industrial growth that creates environmental pollution, urban congestion, and health hazards may increase GNP while simultaneously reducing the quality of life for large segments of the population. The growth criterion also fails to account for the composition of output — an economy may produce more military goods at the expense of healthcare and education, raising GNP without necessarily improving social welfare.

Despite these limitations, the growth criterion highlights the importance of economic efficiency in social welfare analysis. Given that social welfare depends on the amount of goods and services produced (as well as on their distribution), efficiency is a necessary prerequisite for the maximisation of the level of welfare. Economic efficiency can be defined objectively, and modern welfare economics is mainly concerned with the examination (comparison) of the Pareto-efficiency of different economic situations. However, efficiency, although a necessary condition, is not sufficient to guarantee the maximisation of social welfare. Efficiency does not dispose of the need for having an ethical standard of comparing alternative economic states.

(ii) Bentham's Criterion

Jeremy Bentham, an English economist and philosopher, argued that welfare is improved when 'the greatest good (is secured) for the greatest number'. Implicit in this dictum is the assumption that the total welfare is the sum of the utilities of the individuals of the society. The application of Bentham's ethical system to economics has serious shortcomings.

To illustrate the pitfalls, assume that society consists of three individuals, A, B, and C, so that

$$W = U_A + U_B + U_C$$

According to Bentham $\Delta W > 0$ if $(\Delta U_A + \Delta U_B + \Delta U_C) > 0$. However, consider a change where A's and B's utility increases while C's utility decreases, but the sum of the gains to A and B exceeds the loss to C: $(\Delta U_A + \Delta U_B) > |\Delta U_C|$. Under Bentham's criterion, two individuals are better-off while the third is worse-off, but the total utility has increased, so the change is declared a welfare improvement. This obviously implies that A and B have a greater

'worthiness' than C. Thus, implicit in Bentham's criterion is an interpersonal comparison of the deservingness of the members of society — precisely the kind of subjective judgement that welfare economics seeks to avoid.

There is another difficulty with the application of Bentham's criterion: it cannot be applied to compare situations where 'the greatest good' and the 'greatest numbers' do not exist simultaneously. For example, assume that in situation 1, $U_A = 200$, $U_B = 50$, $U_C = 30$, so that total utility is 280. In situation 2, after a redistribution, $U_A = 100$, $U_B = 80$, $U_C = 80$, so that total utility is 260. The first situation has 'the greatest good' ($280 > 260$), but the second involves a more even distribution of a smaller total good among the 'greatest number'. Bentham's criterion provides no basis for choosing between these two fundamentally different configurations.

(iii) Cardinalist Approach

Several economists proposed the use of the 'law of diminishing marginal utility' as a criterion of welfare. Their argument runs as follows. Assume that society consists of three individuals: A has an income of £1,000, while B and C have an income of £500 each. Consumer A can buy double the quantities of goods as compared to B and C. However, given the law of diminishing marginal utility, A's total utility is less than double the total utility of either B or C, because A's marginal utility of money is less than that of B or C. Thus, $W < W^*$ in this situation. To increase social welfare, income should be redistributed from A to B and C. In fact, cardinal welfare theorists would maintain that social welfare would be maximised if income were equally distributed among all members of society.

The cardinalist approach to welfare has a serious flaw: it assumes that all individuals have identical utility functions for money, so that with an equal income distribution, all would have the same marginal utility of money. This assumption is too strong. Individuals differ in their attitudes toward money. A rich person may still derive more utility from additional income due to different preferences, family circumstances, health needs, or entrepreneurial ambitions. Conversely, a poor person may not experience the same increase in utility from additional income if, for instance, their consumption needs are already being met through social programmes.

Furthermore, the cardinalist approach ignores the incentive effects of redistribution. If income is redistributed to achieve equality, the incentive to work, save, and invest may be diminished, potentially reducing the total national income available for distribution. As the textbook notes, excessive redistribution may result in Pareto inefficiency in the use of resources and a reduction in social welfare, even though it achieves a more equal distribution of income.

Question 2: *Pareto optimality is a necessary but not sufficient condition for social welfare maximization' — justify the statement considering a 2×2×2 model.*

The Pareto criterion, named after the Italian economist Vilfredo Pareto (1848–1923), refers to economic efficiency which can be objectively measured. According to this criterion, any change that makes at least one individual better-off and no one worse-off is an improvement in social welfare. Conversely, a change that makes no one better-off and at least one worse-off is a decrease in social welfare. A situation is defined as Pareto-optimal (or Pareto-efficient) if it is impossible to make anyone better-off without making someone worse-off.

In a 2×2×2 model (two factors, two commodities, two consumers), a Pareto-optimal state requires three marginal conditions:

1. **Efficiency in distribution:** $MRS_{x,y}^A = MRS_{x,y}^B$ (equal marginal rate of substitution for all consumers)
2. **Efficiency in production:** $MRTS_{L,K}^x = MRTS_{L,K}^y$ (equal marginal rate of technical substitution for all firms)
3. **Efficiency in product-mix:** $MRPT_{x,y} = MRS_{x,y}^A = MRS_{x,y}^B$ (marginal rate of product transformation equals MRS)

The Three Marginal Conditions for Pareto Optimality in the 2×2×2 Model

In the two-factor (L, K), two-commodity (X, Y), two-consumer (A, B) model, three marginal conditions must be satisfied for a Pareto-efficient allocation:

1. **Efficiency in the distribution of commodities among consumers (efficiency in exchange):** $MRS_{x,y}^A = MRS_{x,y}^B$

This condition requires that the marginal rate of substitution between any two goods be equal for all consumers. It is satisfied at all points on the Edgeworth contract curve of exchange. Points off this curve are Pareto-inefficient because a redistribution can increase at least one consumer's utility without reducing the other's.

2. **Efficiency in the allocation of factors among firms (efficiency in production):** $MRTS_{L,K}^x = MRTS_{L,K}^y$ This condition requires that the marginal rate of technical substitution between any two inputs be equal for all commodities. It is satisfied at all points on the Edgeworth contract curve of production.

3. **Efficiency in the composition of output (product-mix):** $MRPT_{x,y} = MRS_{x,y}^A = MRS_{x,y}^B$. This condition requires that the marginal rate of product transformation between any two commodities be equal to the marginal rate of substitution for any two goods. The MRPT is the slope of the production possibility curve; the MRS is the slope of the indifference curves.

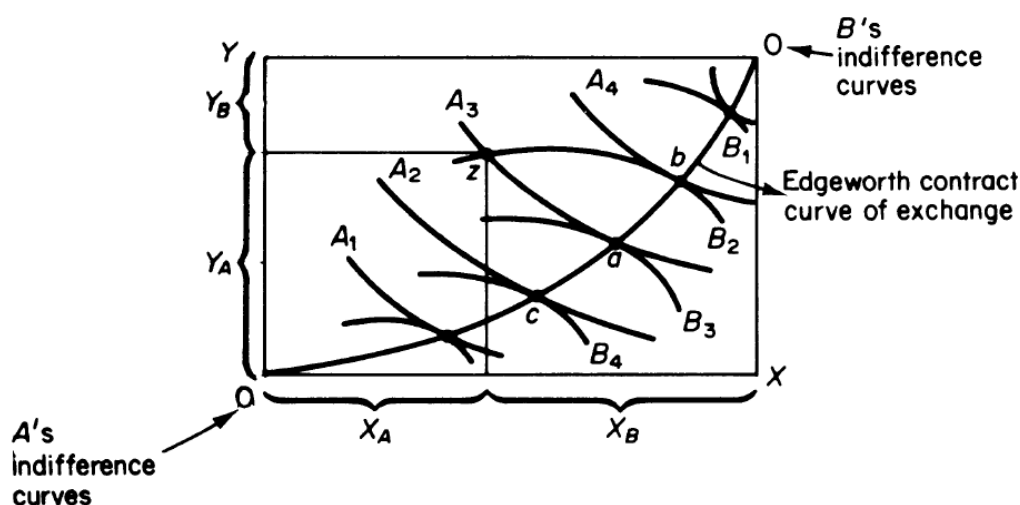


Figure 23.1 Edgeworth box of exchange

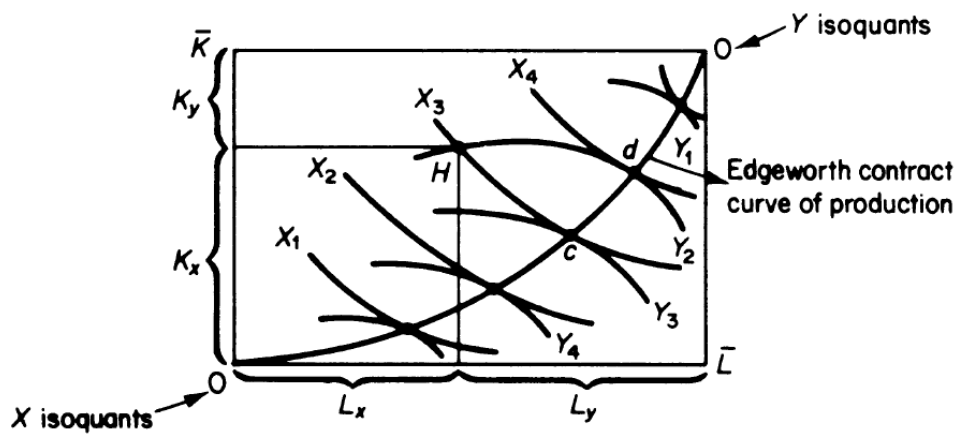


Figure 23.2 Edgeworth box of production

Why Necessary but Not Sufficient

Welfare maximisation occurs at a point on the grand utility possibility frontier (UU'). The grand utility frontier is the envelope of all utility possibility curves, each derived from a different Pareto-efficient product-mix on the production possibility curve. Since every point on the grand utility frontier satisfies all three marginal conditions of Pareto efficiency, any welfare-maximising point must necessarily be Pareto-optimal. This establishes necessity: one cannot maximise social welfare without first achieving Pareto efficiency.

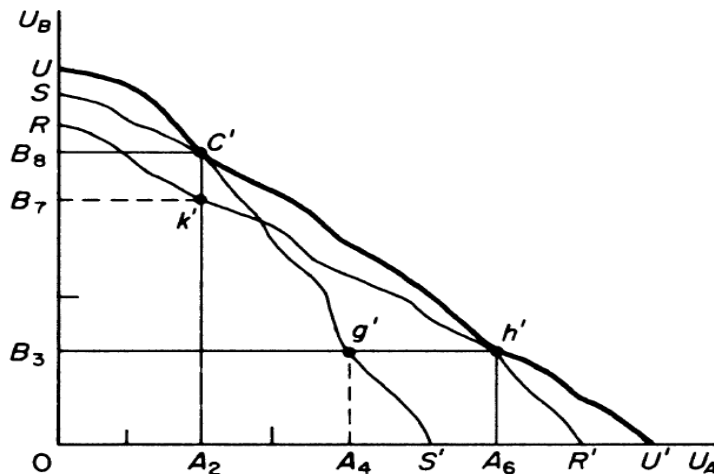


Figure 23.5 The grand utility possibility frontier

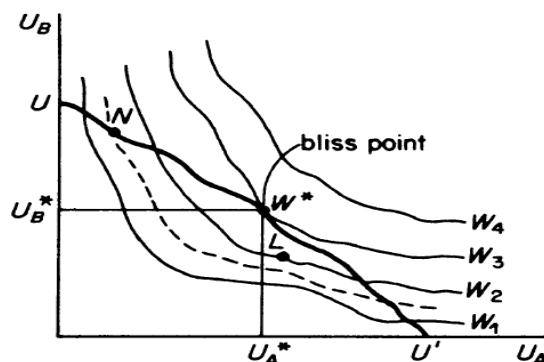


Figure 23.6 Maximisation of social welfare

Figure – The Point of Bliss (Tangency of Social Indifference Contour with Grand Utility Frontier)

There exists a very large number of Pareto-optimal states. All points on the production possibility curve are Pareto-efficient, but they represent vastly different allocations of resources, different output combinations, and different distributions of income between consumers A and B. The Pareto criterion alone cannot rank these alternative Pareto-optimal states because ranking them requires an interpersonal comparison of utility — precisely what the Pareto criterion avoids.

To choose among Pareto-optimal states, a social welfare function is required. Consider a point N on the grand utility frontier that is Pareto-optimal, and a point L that is NOT Pareto-optimal (lying below the grand frontier). As shown in the Figure, point L may lie on a higher social indifference contour than point N, meaning that the non-Pareto-optimal state yields higher social welfare than the Pareto-optimal state. However, it can be shown that given any inefficient point (below the grand utility frontier), there will exist some point(s) on the grand utility frontier that represents an improvement in social welfare over that inefficient point.

The point of maximum social welfare — the 'bliss point' w^* — occurs where the grand utility possibility frontier is tangent to the highest attainable social indifference contour. At this point, and only at this point, all three Pareto conditions are satisfied AND social welfare is maximised.

Question 3: *Explain the underlying assumption about the marginal utility of money for the Hicks-Kaldor compensation criterion to be correct.*

Nicholas Kaldor and John Hicks suggested an approach to evaluating welfare changes that involves the concept of hypothetical compensation. Assume that a change in the economy is being considered, which will benefit some ('gainers') and hurt others ('losers'). One asks the 'gainers' how much money they would be prepared to pay in order to have the change implemented, and the 'losers' how much money they would be prepared to pay in order to prevent the change. If the amount of money of the 'gainers' is greater than the amount of the 'losers', the change constitutes an improvement in social welfare, because the 'gainers' could compensate the 'losers' and still be left with some 'net gain'. Thus, the criterion states: a change constitutes an improvement in social welfare if those who benefit from it could compensate those who are hurt, and still be left with some 'net gain'.

The Critical Assumption — Equal Marginal Utility of Money

The Kaldor-Hicks criterion evaluates alternative situations on the basis of monetary valuations of different persons. In doing so, it implicitly assumes that the marginal utility of money is the same for all individuals in the society. The logic is as follows: when comparing the amount that gainers are willing to pay with the amount that losers would require as compensation, we are effectively comparing utils (units of utility). For monetary amounts to be comparable across individuals in utility terms, each unit of money must represent the same amount of utility for each person. If the marginal utility of money differs across individuals, then a gain of £1,000 to a millionaire represents far fewer utils than a loss of £1,000 to a poor person, and the comparison becomes meaningless.

Why This Assumption Is Unrealistic

In the real world, income distribution is highly unequal, and the marginal utility of money varies significantly across individuals. Consider an economy with two individuals: A, who is a millionaire, and B, who has an income of £4,000. Suppose the government considers a change that will benefit A and hurt B. A is willing to pay £2,000 for this change, while B would pay £1,000 to prevent it. According to the Kaldor-Hicks criterion, the change improves welfare (net gain of £1,000 after hypothetical compensation). However, the gain of £2,000 provides very little additional utility to the millionaire A, whose marginal utility of money is extremely low, while the loss of £1,000 represents a substantial reduction in well-being for B, whose marginal utility of money is very high. The total welfare may actually decrease, not increase, if the change takes place.

Additional Criticisms

The Kaldor-Hicks criterion also suffers from the Scitovsky paradox: it is possible for situation K to be declared superior to situation J by the Kaldor criterion, while simultaneously J is declared superior to K by the same criterion. This intransitivity arises because the distribution of income differs between the two situations, affecting the marginal utility of money and hence the willingness to pay or accept compensation.

Furthermore, the criterion is based on hypothetical compensation, not actual compensation. The 'gainers' are not required to actually compensate the 'losers'; the test is purely hypothetical. This means that the criterion can declare a change a welfare improvement even when the losers are made significantly worse-off in reality. The criterion ignores the existing income distribution and makes implicit interpersonal comparisons by assuming that the same amounts of money have the same utility for individuals with vastly different incomes. Only if the marginal utility of money were equal for all individuals would the Kaldor-Hicks criterion be a 'correct' welfare measure — but this condition can only hold under perfectly equal income distribution.

Question 4: Define social welfare function. How could it be used as a criterion to measure social welfare? Explain.

The social welfare function (SWF) was introduced by Abram Bergson as an explicit set of ethical value judgements that permits a unique preference-ordering of all possible economic states (configurations) in which different individuals enjoy different utility levels. If the economy consists of two individuals A and B, the social welfare function can be written as

$$W = f(U_A, U_B)$$

where U_A and U_B are the utility levels (in ordinal indexes) of consumers A and B, respectively.

Graphical Representation

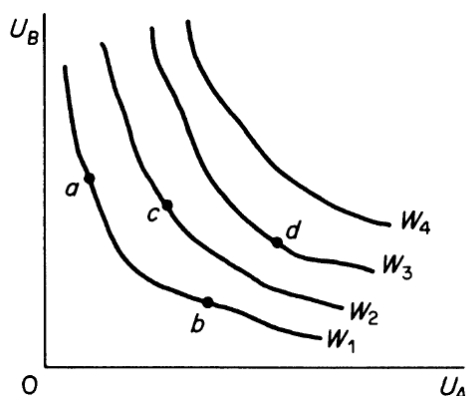


Figure – Bergson's Welfare Contours (Social Indifference Curves in Utility Space)

The social welfare function can be represented by a set of social indifference contours (also called Bergson welfare contours) in utility space. On a graph whose axes measure the utilities of A and B, each social indifference contour is the locus of combinations of (U_A, U_B) that yield the same level of social welfare. The contours have the standard properties of indifference curves: they are downward-sloping (more utility for one individual must be offset by less for the other to maintain the same welfare level), they do not intersect, and higher contours (further to the northeast) represent higher levels of social welfare.

With such a set of social indifference contours, alternative economic states can be unambiguously evaluated. For example, a change which would move the society from point b to point c (or d) increases social welfare. A change moving the society from point a to point b leaves the level of social welfare unaltered.

Use as a Criterion for Measuring Social Welfare

The social welfare function serves as a comprehensive welfare criterion when used in conjunction with the grand utility possibility frontier. The procedure is as follows:

- Step 1:** Derive the grand utility possibility frontier, which shows the maximum utility attainable by one consumer given the utility of the other, across all possible product-mixes and efficient distributions.
- Step 2:** Superimpose the social indifference contours on the grand utility frontier.
- Step 3:** The social welfare-maximising point is the point of tangency between the grand utility frontier and the highest attainable social indifference contour. This point is called the 'bliss point'.

The social welfare function thus provides a complete and unambiguous method for selecting among all Pareto-optimal states. While the Pareto criterion can identify efficient states, it cannot rank them. The SWF fills this gap by incorporating ethical judgements about the relative deservingness of individuals into the ranking mechanism.

Determination of Welfare-Maximising Variables

At the bliss point, the welfare-maximising values of all ten unknowns in the $2 \times 2 \times 2$ model are determined uniquely: the total quantities of the two products (X^* and Y^*), the quantities consumed by each consumer ($X_A^*, X_B^*, Y_A^*, Y_B^*$), and the allocation of resources between industries ($L_x^*, L_y^*, K_x^*, K_y^*$). The bliss point is associated with a unique commodity-mix on the PPC, a unique distribution of commodities on the Edgeworth contract curve of exchange, and a unique allocation of factors on the Edgeworth contract curve of production.

The Problem of Construction

The fundamental limitation of the social welfare function is that there is no objective method of constructing it. Its existence is axiomatically assumed in welfare economics. The SWF requires somebody to compare various individuals or groups and rank them according to what they think their worthiness is — an inherently subjective exercise involving value judgements. A democratically elected government could be assumed to make such value judgements acceptable to the society as a whole, but this remains an implicit assumption.

Arrow's Impossibility Theorem demonstrates that no social welfare function can be constructed from individual preference orderings that simultaneously satisfies a set of reasonable conditions (unrestricted domain, Pareto principle, independence of irrelevant alternatives, and non-dictatorship). This further underscores the difficulty of constructing a consistent and democratic SWF.

It should also be noted that the social welfare function cannot be used to derive community indifference curves in output space without taking into account the distribution of income among individuals. The community MRS at a given point in commodity space depends on how the goods are distributed among individuals, i.e., on which utility combination on the Edgeworth contract curve of exchange is chosen.

Question 5: *What are the major assumptions of social welfare maximization in a $2 \times 2 \times 2$ model?*

The social welfare maximisation problem in the $2 \times 2 \times 2$ model is formulated under the following set of assumptions, each of which plays a critical role in ensuring that the model yields a determinate solution:

1. There are two factors, labour L , and capital K , whose quantities are given (in perfectly inelastic supply). These factors are homogeneous and perfectly divisible.
2. Two products, X and Y , are produced by two firms. Each firm produces only one commodity. The production functions give rise to smooth isoquants, convex to the origin, with constant returns to scale. Indivisibilities in the production processes are ruled out.
3. There are two consumers whose preferences are represented by indifference curves, which are continuous, convex to the origin and do not intersect.
4. The goal of consumers is utility maximisation and the goal of firms is profit maximisation.
5. The production functions are independent. This rules out joint products and external economies and diseconomies in production.
6. The utilities of consumers are independent. Bandwagon, snob and Veblen effects are ruled out. There are no external economies or diseconomies in consumption.
7. The ownership of factors, that is, the distribution of the given L and K between the two consumers, is exogenously determined.
8. A social welfare function, $W = f(U_A, U_B)$, exists. This permits a unique preference-ordering of all possible states, based on the positions of the two consumers in their own preference maps. This welfare function incorporates an ethical valuation of the relative deservingness or worthiness of the two consumers.

The problem is to determine the welfare-maximizing values of ten variables: the total quantities of X and Y , their distribution between the two consumers (X_A, X_B, Y_A, Y_B), and the welfare-maximising allocation of resources (L_x, L_y, K_x, K_y).

Question 6: Derive the grand utility possibility frontier and determine the 'point of bliss.'

Derivation of the Grand Utility Possibility Frontier

Step 1: Start with the production possibility curve (PPC), which is the locus of all Pareto-efficient output combinations (X, Y) derived from the Edgeworth contract curve of production.

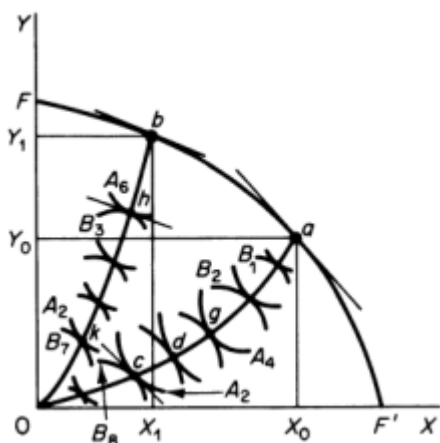


Figure – Product-mixes on the PPC and Edgeworth Boxes of Exchange

Step 2: For each product-mix on the PPC (e.g., Y_0, X_0 denoted by point a), construct the Edgeworth box of exchange. Points on the contract curve of exchange (0a) represent Pareto-optimal distributions of that product-mix between consumers A and B.

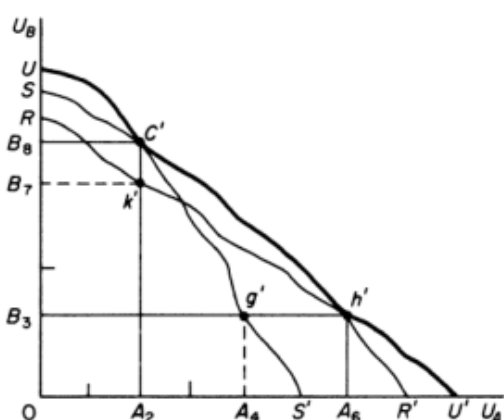


Figure – The Grand Utility Possibility Frontier (Envelope of Utility Frontiers)

Step 3: Each point on the contract curve corresponds to a different utility combination (U_A, U_B). Map all such utility combinations from contract curve 0a into utility space to obtain the utility possibility frontier SS' for the specific product-mix (Y_0, X_0).

Step 4: Using the third Pareto condition ($MRPT_{x,y} = MRS_{x,y}^A = MRS_{x,y}^B$), for each product-mix on the PPC, identify the single point on the corresponding utility frontier where $MRPT = MRS$. This gives the "grand" utility for that product-mix.

Step 5: The envelope of all such points (one from each utility possibility frontier) forms the grand utility possibility frontier (UU'). It shows the maximum utility attainable by one consumer given the utility of the other, across all possible product-mixes and optimal distributions.

In summary, each point on the PPC gives rise to a utility possibility frontier. The envelope of these utility possibility frontiers is the grand utility possibility frontier of the economy.

Alternative Derivation

Using the third marginal condition ($MRPT = MRS$), for each point on the PPC, pick the point on the corresponding contract curve with the same slope as the PPC. From the utilities at this point, plot a single point in utility space. Repeating for all PPC points yields the envelope frontier.

Thus, all points on the grand utility possibility frontier satisfy all the Pareto marginal conditions of efficiency: efficiency in production, efficiency in distribution, efficiency in product composition.

Determination of the Point of Bliss

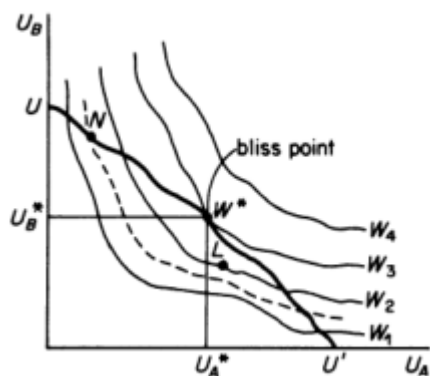


Figure – The Point of Bliss (Tangency of Social Indifference Contour with Grand Utility Frontier)

Social welfare is maximised at the point of tangency of the grand utility possibility frontier UU' with the highest possible social indifference contour. This tangency point is called the 'point of bliss', denoted w^* . At w^* , the maximum social welfare attainable in the economy is the level implied by the indifference contour W_3 . The two consumers enjoy utility levels U_A^* and U_B^* respectively.

From the bliss point, all ten unknowns of the $2 \times 2 \times 2$ model can be determined uniquely by retracing the derivation steps: (1) w^* corresponds to a unique utility combination (U_A^*, U_B^*) ; (2) This utility combination belongs to a specific utility possibility frontier; (3) That utility frontier corresponds to a unique point on the PPC — this defines the welfare-maximising commodity-mix (X^*, Y^*) ; (4) The Edgeworth box of exchange for (X^*, Y^*) contains the specific distribution point that yields (U_A^*, U_B^*) — this defines $(X_A^*, X_B^*, Y_A^*, Y_B^*)$; (5) The PPC point maps back to the Edgeworth box of production, defining the optimal resource allocation $((L_x^*, L_y^*, K_x^*, K_y^*))$.

Question 7: Critically explain the three key things that Pigou dealt with in analyzing social welfare.

A.C. Pigou, in his seminal work 'The Economics of Welfare' (1920), laid the foundations of welfare economics by identifying three fundamental dimensions of social welfare analysis. His three contributions — efficiency, equity through redistribution, and correcting externalities — remain the pillars of welfare economic analysis. The three key aspects that Pigou dealt with in analyzing social welfare are:

1. The Size of the National Dividend (Economic Efficiency)

Pigou's first concern was with the aggregate size of national output. He argued that social welfare depends fundamentally on the total quantity of goods and services available for consumption — the 'national dividend' in Pigou's terminology. A larger national dividend means that, in principle, more goods and services are available to satisfy the needs and wants of society's members. This aligns with the GNP growth criterion. Pigou emphasised that increasing the total output of the economy is a prerequisite for improving social welfare. In the language of the 2×2 model, this means moving the economy to the production possibility frontier and operating at a point where all resources are efficiently employed. The efficiency dimension is captured by the three marginal conditions of Pareto optimality: efficiency in exchange, efficiency in production, and efficiency in the product-mix.

However, Pigou recognised that efficiency alone is not sufficient. The size of the national dividend tells us nothing about how the output is distributed among members of society. Two economies with the same GNP can have vastly different levels of social welfare depending on how income is distributed.

2. The Distribution of the National Dividend (Equity through Redistribution)

Pigou's second concern was with the equitable distribution of income. He argued that, given the law of diminishing marginal utility, redistributing income from the rich to the poor would increase total social welfare even if total output remained unchanged. This is because the marginal utility of an additional pound of income is higher for a poor person than for a rich person. This is the cardinalist approach to welfare. Cardinal welfare theorists maintained that social welfare would be maximised if income were equally distributed among all members of society, since equal distribution would equalise the marginal utility of income across all individuals.

Pigou's analysis of distribution directly connects to the social welfare function. The choice of a particular point on the grand utility possibility frontier — the selection of the bliss point — inherently involves an ethical judgement about the distribution of welfare between individuals A and B. Different social welfare functions, reflecting different ethical values, would select different bliss points and hence different distributions of income and output.

3. Externalities and the Divergence Between Private and Social Costs/Benefits (Correcting Externalities)

Pigou's third major contribution was his analysis of externalities — situations where the actions of an individual economic agent create costs or benefits for others that are not reflected in market prices.

When private cost is less than social cost, the condition for optimal resource allocation is violated. For example, consider a firm producing alcohol in a perfectly competitive market. The firm is in equilibrium when $MC_x = P_x$, where MC_x is the private marginal cost. This does not include the cost of pollution, health hazards, and accidents caused by alcohol consumption. The marginal social cost is: $MSC_x = MC_x + MEC$, where MEC is the marginal external cost. Since $MC_x = P_x$, it follows that $P_x < MSC_x$, which implies that the allocation of resources to the production of alcohol is not socially optimal. The firm produces too much because it does not bear the full social cost of its production.

Pigou's solution was to impose a tax (a 'Pigouvian tax') equal to the marginal external cost, so that the firm's effective marginal cost becomes equal to the social marginal cost. The firm would then produce where: $P_x = MSC_x = MC_x + MEC$. This is the socially optimal level of output. Similarly, when there are external benefits

($MSB > P$), the socially optimal output is larger than the private equilibrium output, and Pigou recommended subsidies to correct the underproduction.

The general condition for optimal resource allocation in a multi-product economy with externalities: $\frac{MSB_x}{MSC_x} = \frac{MSB_y}{MSC_y} = \dots = \frac{MSB_M}{MSC_M}$. Without externalities, $MSB = P = MSC$, and this condition reduces to the standard competitive equilibrium condition.

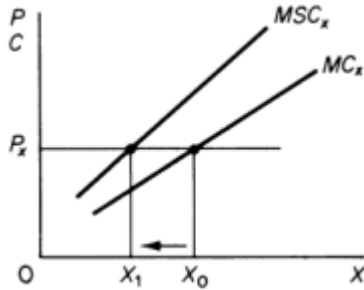


Figure – $MSC > MC$: Overproduction Due to External Cost

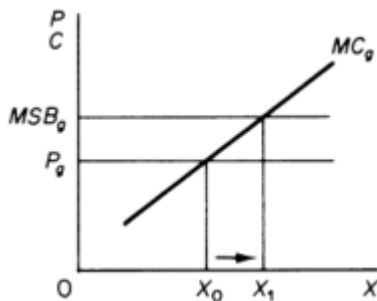


Figure – $MSB > P$: Underproduction Due to External Benefit

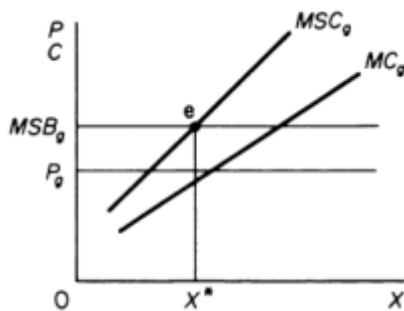


Figure – Equilibrium Where $MSC = MSB$

Critical Evaluation

While Pigou's framework was pioneering, it has limitations. The cardinalist approach to income distribution assumes identical utility functions, which is empirically implausible. The externality analysis assumes that the government can accurately measure marginal external costs and benefits — an assumption that is often violated in practice. Nevertheless, Pigou's three pillars — efficiency, equity, and externality correction — remain the foundation of modern welfare economics.

Question 8: Does perfect competition ensure welfare maximization? Examine.

Under the assumptions of the 2×2×2 model, perfect competition does ensure the attainment of maximum social welfare. This is the central result of modern welfare economics, often referred to as the First Welfare Theorem. The theorem states that a perfectly competitive equilibrium is Pareto-optimal. Combined with the existence of a social welfare function, this implies that the perfectly competitive equilibrium coincides with the point of maximum social welfare — the bliss point.

Proof in the 2×2×2 Model

(a) Efficiency in factor substitution (Production equilibrium)

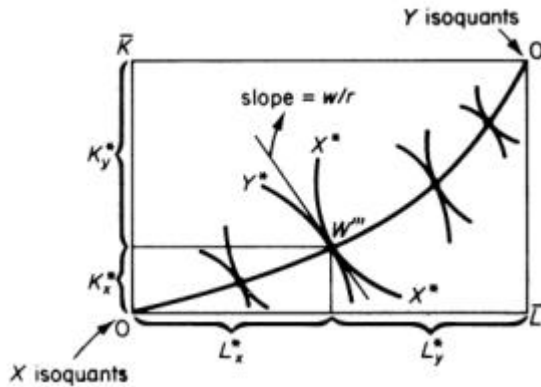


Figure – Optimal Allocation of Resources K^* and L^*

Profit maximisation by the individual firm requires least-cost production of the profit-maximising output. The firm minimises cost by choosing the input combination where the marginal rate of technical substitution equals the factor price ratio:

$$MRTS_{L,K}^x = \frac{w}{r}$$

Since in perfect competition all firms face the same factor prices, it follows that:

$$MRTS_{L,K}^x = MRTS_{L,K}^y = \frac{w}{r}$$

This is the first marginal condition for Pareto optimality. In terms of the Edgeworth box of production, the common MRTS defines the point on the contract curve where the slope of the isoquants equals the slope of the isocost line.

Since all firms face the same factor prices, the MRTS is equalized across firms. This is the first Pareto condition.

(b) Efficiency in distribution (Consumption equilibrium)

Utility maximisation by each consumer requires that the consumer choose the product-mix where the marginal rate of substitution equals the commodity price ratio:

$$MRS_{x,y}^A = \frac{P_x}{P_y}$$

Since in perfect competition all consumers face the same commodity prices, it follows that:

$$MRS_{x,y}^A = \frac{P_x}{P_y} = MRS_{x,y}^B$$

Since all consumers face the same commodity prices, the MRS is equalized across consumers. This is the second Pareto condition.

(c) Efficiency in product-mix (Simultaneous equilibrium)

At the bliss point, the equalised slope of the indifference curves (the common MRS) equals the slope of the production possibility curve at the optimal product-mix. Since $MRPT_{x,y} = \frac{MC_x}{MC_y}$ and firms set $P = MC$ in perfect competition, we have:

$$MRPT_{x,y} = \frac{P_x}{P_y} = MRS_{x,y}^A = MRS_{x,y}^B$$

This is the third Pareto condition.

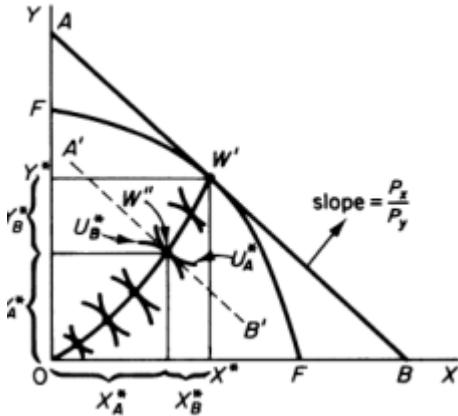


Figure – Determination of Welfare-Maximising Outputs (W') and Distribution (W'')

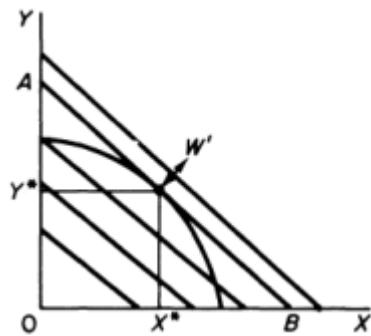


Figure – Isovalue Line Tangent to PPC at the Welfare-Maximising Product-Mix

The "duality theorem" of welfare economics states that welfare maximization is independent of prices. However, implicit in the purely technocratic formulation is a set of constants that can be the prices of a perfectly competitive economy. When individual profit maximizers and utility maximizers act in response to these prices, their behavior leads to social welfare maximization. Adam Smith's "invisible hand" proposition is thus formalized: each individual, pursuing self-interest, is led to promote the general welfare.

However, perfect competition is only one way of attaining welfare maximization. A decentralized socialist system (Lerner-Lange type), where the government estimates "shadow prices" and directs economic units to maximize their gains, can in principle achieve the same results. The model collapses when its assumptions are violated (externalities, increasing returns, indivisibilities, absence of social welfare function).

Question 9: Graphically explain the problems of existence, uniqueness and stability in connection to general equilibrium.

Three fundamental problems arise in connection with general equilibrium:

1. Existence Problem

A general equilibrium solution exists if each market is cleared at a positive price, with each consumer maximising satisfaction and each firm maximising profit. In the partial-equilibrium context, an equilibrium exists when the demand and supply curves intersect at a positive price where $Q_D = Q_S$.

The excess demand function $E(P) = Q_{D(P)} - Q_{S(P)}$ provides a powerful analytical tool. An equilibrium exists when $E(P) = 0$, i.e., when the excess demand curve intersects the vertical price-axis. If $E(P)$ lies entirely above the axis at all prices, excess demand is always positive and no equilibrium price exists (the market cannot clear). If $E(P)$ lies entirely below the axis, excess supply always prevails.

In general equilibrium, Arrow and Debreu (1954) provided a rigorous proof of existence for perfectly competitive markets with no indivisibilities and no increasing returns to scale. The proof relies on fixed-point theorems (Brouwer's or Kakutani's) rather than simple equation counting. Arrow and Hahn (1971) extended the proof to economies with limited increasing returns and monopolistic competition, without indivisibilities. The key takeaway from the existence literature is that the mere equality of the number of equations and unknowns in the Walrasian system is neither necessary nor sufficient for existence. The existence proof requires well-behaved (continuous, convex) production and demand functions.

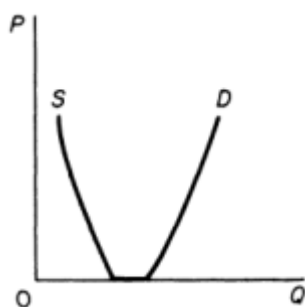


Figure – No Equilibrium Exists (Demand and Supply Do Not Intersect)

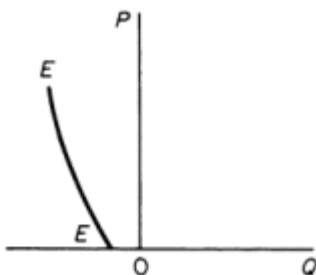


Figure – No Equilibrium Exists in Excess Demand Function

2. Uniqueness Problem

If an equilibrium exists, it may not be unique. Multiple equilibria arise when the excess demand function intersects the vertical price-axis more than once. In this case, there are several distinct price-quantity combinations at which markets clear. The number of equilibria equals the number of intersections of $E(P)$ with the price-axis. In a partial equilibrium context, multiple equilibria can arise from non-linear demand and supply functions that cross more than once. In general equilibrium, multiple equilibria are more common because of the interdependence among markets.

Uniqueness can be guaranteed under certain conditions. If all commodities are gross substitutes (i.e., the cross-price effect on excess demand is positive), then the equilibrium is unique. Uniqueness can also be guaranteed if the weak axiom of revealed preference holds in the aggregate.

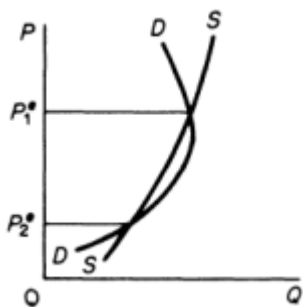


Figure – Multiple Equilibria

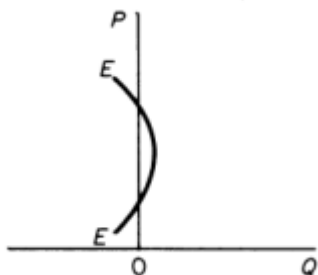


Figure – Multiple Equilibria in Excess Demand Space

3. Stability Problem

If an equilibrium exists, it may be stable or unstable. An equilibrium is stable if, following a disturbance, the system converges back to the equilibrium. Stability depends on the dynamic adjustment process — the mechanism by which prices change in response to excess demand or supply.

In terms of the excess demand function, stability requires that the slope of $E(P)$ at the equilibrium point be negative. When the slope is negative: if price is above equilibrium, $E(P) < 0$ (excess supply), which pushes price down; if price is below equilibrium, $E(P) > 0$ (excess demand), which pushes price up. This self-correcting mechanism drives the system back to equilibrium. When the slope is positive: if price is above equilibrium, $E(P) > 0$ (excess demand), which pushes price further up; if price is below equilibrium, $E(P) < 0$ (excess supply), which pushes price further down. The system diverges from equilibrium — the equilibrium is unstable.

Walras proposed the tâtonnement (groping) process as the adjustment mechanism: an external auctioneer adjusts prices iteratively, increasing prices in markets with excess demand and decreasing prices in markets with excess supply. Walras argued that the system would eventually converge because supply and demand respond more to own-price changes than to other-price changes. However, the tâtonnement process is unsatisfactory for several reasons: it is essentially timeless (no transactions occur until equilibrium is reached), it evades the real-time adjustment issue, and there is no guarantee of convergence in all cases. The textbook discusses two important stability theorems: (a) if all commodities are strict gross substitutes, the equilibrium is stable; and (b) if the weak axiom of revealed preference holds in the aggregate, the system is globally stable.

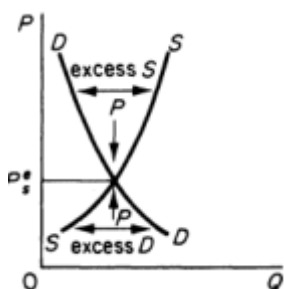


Figure – Unique, Stable Equilibrium

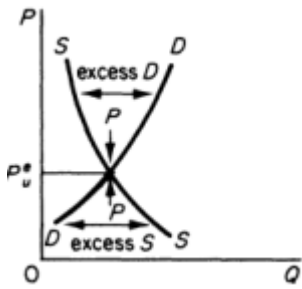


Figure – Unique, Unstable Equilibrium

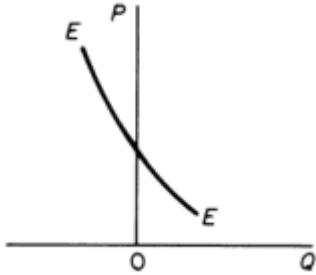


Figure – Stable Equilibrium: Negative Slope of $E_{(P)}$

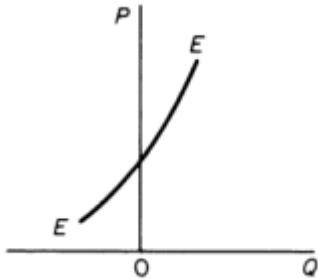


Figure – Stable Equilibrium: Positive Slope of $E_{(P)}$

Question 10: Define consumption externality. Explain by Bandwagon effect and Snob effect.

A consumption externality exists when the utility level (satisfaction) of one consumer is directly affected by the consumption pattern of another consumer. In the standard $2 \times 2 \times 2$ model, it is assumed that consumer choices are independent — the consumption patterns of A do not affect B's utility, and vice versa. When this assumption of independence is violated, the standard conditions for Pareto optimality — particularly the equalisation of the marginal rate of substitution across consumers — do not ensure Pareto-optimal outcomes. The textbook explicitly states that when externalities in consumption are present, the equalisation of the marginal rate of substitution of commodities among consumers does not lead to Pareto optimality. This is because the private marginal rate of substitution no longer reflects the true social marginal rate of substitution when one consumer's utility depends on another's consumption.

Bandwagon Effect

The bandwagon effect occurs when a consumer's demand for a commodity increases because other consumers are also purchasing the same commodity. This is an imitation or conformity effect driven by fashion trends, social pressure, or the desire to belong to a group. When a new style of clothing becomes popular, for example, individuals may purchase it not because of its intrinsic utility but because they observe others wearing it and wish to conform. In terms of the formal model, the bandwagon effect means that consumer A's utility function depends positively on consumer B's consumption of good X. If A's indifference map is drawn in (X, Y) space, a change in B's consumption of X shifts A's entire indifference map — specifically, an increase in B's consumption of X shifts A's indifference curves outward (representing a positive externality in consumption). This creates an external economy for consumer A arising from B's consumption. Economically, the bandwagon effect means that the market demand curve for the commodity is more elastic than it would be in the absence of the externality. A small price decrease triggers a larger quantity increase because the initial purchasers attract additional purchasers through the imitation effect. This can lead to instability in the market and may result in overconsumption relative to the social optimum.

Snob Effect

The snob effect is the reverse of the bandwagon effect: a consumer's demand for a commodity decreases because others are also purchasing it. Snobs derive utility from exclusivity and uniqueness — they value goods precisely because few others possess them. If a luxury item (such as a designer handbag or a rare artwork) becomes widely adopted, a snob consumer's valuation of it decreases because it no longer serves as an effective status signal. In the formal model, consumer A's utility function depends negatively on consumer B's consumption of good X. An increase in B's consumption of X shifts A's indifference curves inward (representing a negative externality — an external diseconomy in consumption). The market demand curve under the snob effect is less elastic than it would be otherwise, because increased popularity of the good reduces the quantity demanded by snob consumers.

Implication for Welfare

When consumption externalities are present, the standard first-order conditions for Pareto optimality are no longer valid. The textbook illustrates this with a detailed example using following Figures. Suppose there is a consumption externality in commodity X (for example, tobacco smoking): A's consumption of X creates a negative externality for B. Even though at the initial competitive equilibrium $MRS_{x,y}^A = MRS_{x,y}^B$, a redistribution of the same total output can increase welfare. If A's consumption of X is reduced, B's indifference map shifts upward (B's utility increases because of reduced exposure to the negative externality). In the new equilibrium, A remains on the same indifference curve, while B moves to a higher indifference curve — total welfare has increased despite the fact that the MRS of the two consumers is no longer equal. This demonstrates that the competitive equilibrium with consumption externalities is not Pareto-optimal, and that the standard condition of MRS equalisation is neither necessary nor sufficient for Pareto efficiency when utility functions are interdependent.

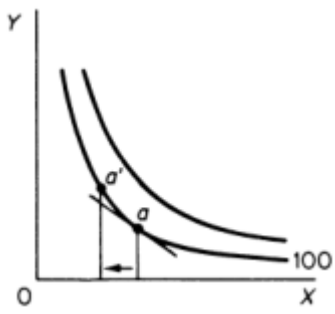


Figure – Consumer A's Indifference Map (with External Effects)

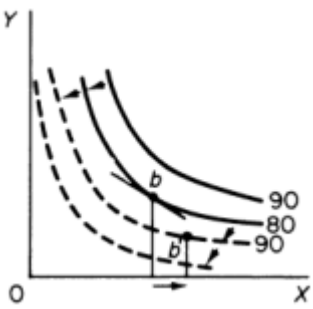


Figure – Consumer B's Indifference Map (shifted due to External Effects)

Question 11: Does consumption externality affect the general equilibrium solution in a $2 \times 2 \times 2$ model?

Yes, consumption externality significantly affects the general equilibrium solution in a $2 \times 2 \times 2$ model. The standard model assumes independent utility functions (Assumption 6), meaning the consumption patterns of A do not affect B's utility, and vice versa. When this assumption is violated by consumption externalities, the model's results change fundamentally.

Specific Effects

- Failure of the MRS equalization condition:** The standard general equilibrium solution requires three marginal conditions to be satisfied simultaneously. The second condition — efficiency in exchange — requires that

$$MRS_{x,y}^A = MRS_{x,y}^B$$

When consumption externalities exist, this condition no longer ensures Pareto optimality. The equalisation of private MRS does not account for the social effects of one consumer's consumption on the other's utility. The social MRS of consumer A differs from the private MRS by the marginal external effect of A's consumption on B's utility. Formally, if A's utility depends on B's consumption of X, then: $Social\ MRS_A = Private\ MRS_A + \left(\frac{\partial U_B}{\partial X_A}\right) \cdot \left(\frac{\partial X_A}{\partial Y_A}\right)$. The competitive equilibrium equates only the private MRS values, ignoring the external term. The resulting allocation is not Pareto-optimal.

- Inefficiency of competitive equilibrium:** Even if all three standard marginal conditions appear to be satisfied simultaneously, the presence of consumption externalities means the economy is not truly at a Pareto-optimal point. Look at the following figures. Consider a competitive equilibrium where A consumes tobacco (good X) and the consumption of tobacco creates a negative externality for B. At the competitive equilibrium, $MRS_A = MRS_B$. However, if A's consumption of X is reduced (and the released X is redistributed), B's indifference map shifts upward due to the reduction in the negative externality. The new allocation has A on the same indifference curve and B on a higher one — proving that the original equilibrium was NOT Pareto-optimal.

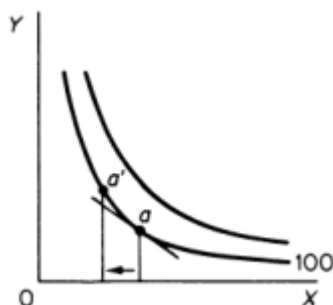


Figure – Consumer A's Indifference Map (with External Effects)

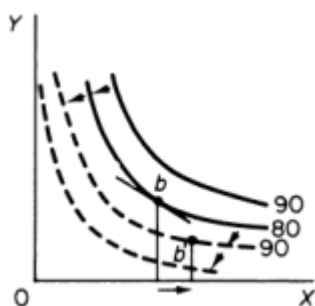


Figure – Consumer B's Indifference Map (shifted due to External Effects)

- Divergence Between Private and Social Valuation:** If externalities exist, the model breaks down for two reasons: first, the Pareto optimality conditions are violated; second, there is a difference between the private cost (or benefit) of an action to the society as a whole.' In other words, externalities create a divergence between private and social costs and benefits. Because externalities are not reflected in market prices, these prices provide misleading information (signals) for an optimal allocation of resources. Consumers base their

decisions on private valuations (private MRS) rather than social valuations (social MRS), leading to a misallocation of resources.

4. **Redistribution can improve welfare:** When consumption externalities exist, redistributing the same total output of goods between consumers can increase social welfare in multiple ways. One consumer remains on the same indifference curve while the other moves to a higher indifference curve due to the externality. The direction of the welfare-improving redistribution depends on the sign of the externality: for negative externalities, reducing the consuming individual's quantity of the externality-generating good improves welfare; for positive externalities, increasing it may improve welfare.

Question 12: Graphically explain how an external shock in the product market affects the factor market and the mechanism through which equilibrium is restored in both markets.

Initial Setup

Consider a $2 \times 2 \times 2$ economy with the following initial conditions: two substitute commodities, X and Y, produced in perfectly competitive industries; two factors of production, capital K and labour L, in fixed supply with perfectly competitive factor markets; production functions are continuous with diminishing MRTS and decreasing returns to scale; Industry X is less capital-intensive than Y: $\left(\frac{K}{L}\right)_x < \left(\frac{K}{L}\right)_y$; consumers maximise utility; firms maximise profit; the system is initially in simultaneous equilibrium in all four markets (X, Y, L, K).

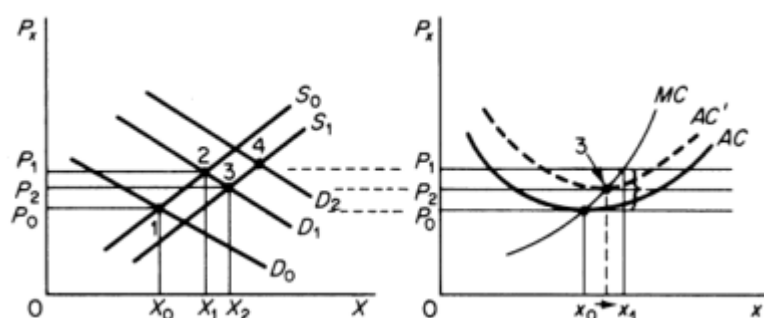


Figure – Left Panel: Industry X: Demand Shift and Price Increase; Right Panel: A Firm in Industry X: Short-Run Profit

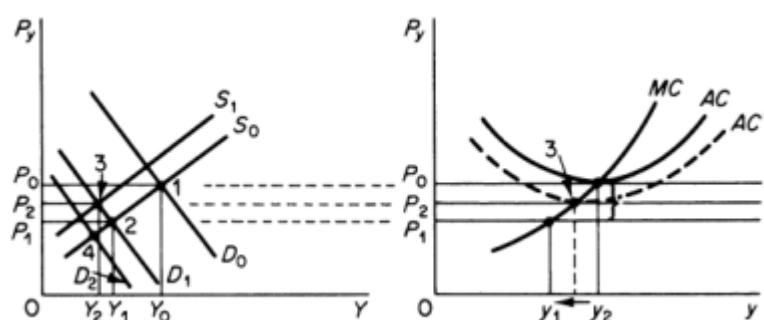


Figure – Left Panel: Industry Y: Demand Shift and Price Decrease; Right Panel: A Firm in Industry Y: Short-Run Loss

Step 1 — External Shock in the Product Market

An exogenous change in consumer tastes shifts the demand for X rightward from D_0 to D_1 . This is the initial disturbance that sets in motion the chain of adjustments across all markets. In the short run: Price of X rises from P_0 to P_1 ; Quantity of X sold increases from X_0 to X_1 . Since X and Y are substitutes (they cannot be complementary in the $2 \times 2 \times 2$ model), the increase in demand for X is accompanied by a decrease in demand for Y. Demand for Y shifts leftward; P_Y falls and quantity of Y decreases from Y_0 to Y_1 .

Step 2 — Firm-Level Response in Producer Markets

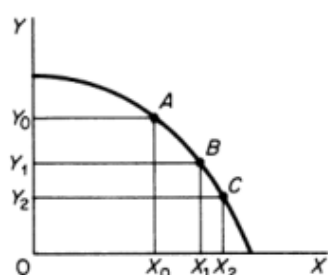


Figure – Movement Along the Production Possibility Curve from A to B

The increase in P_x creates excess profits for producers of X ($P_1 > AC$ at the original output). The decrease in P_y creates losses for producers of Y ($P_1 < AC$ at the original output). In the short run, each firm in X expands output along its MC curve to the point where $MC = P_1$. Each firm in Y contracts output. In the long run, excess profits attract entry of new firms in industry X, while losses induce exit of firms from industry Y. The economy moves from point A to point B on the production possibility curve, reflecting the reallocation of resources from Y to X.

Step 3 — Effect on Factor Markets (Disequilibrium)

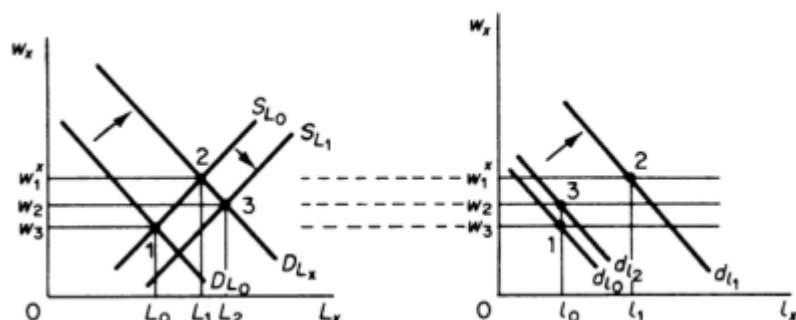


Figure – Left Panel: Labour Market for Industry X; Right Panel: Demand for Labour by a firm in industry X

The expansion of industry X and entry of new firms increase the demand for both labour and capital in industry X. The demand curves D_{L_x} and D_{K_x} shift outward. The wage rate w_x and the rental rate r_x in industry X rise.

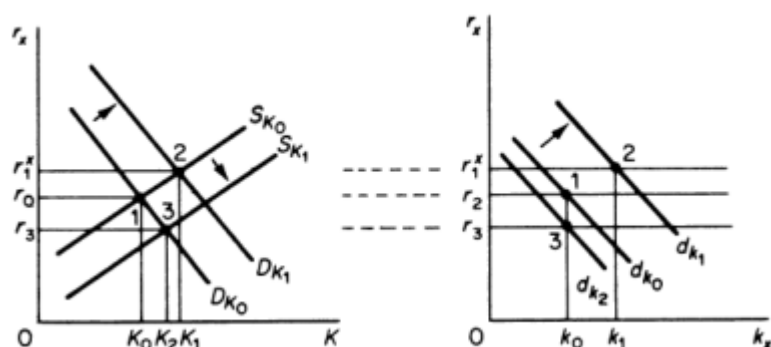


Figure – Left Panel: Market for Capital in Industry X; Right Panel: Demand for Capital by a firm in industry X

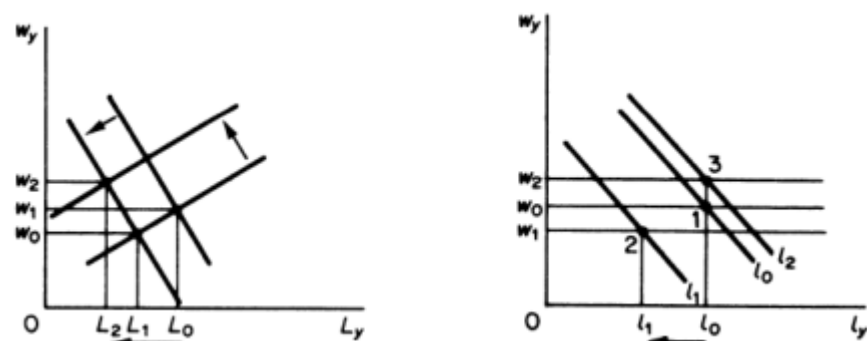


Figure – Left Panel: Market for Labour in Industry Y; Right Panel: Demand for Labour by a firm in industry Y

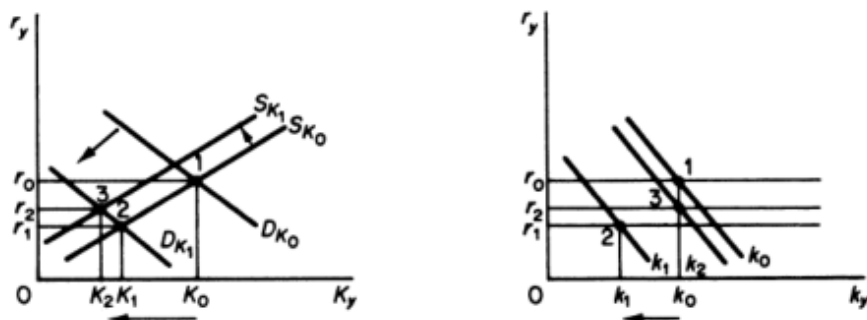


Figure – Left Panel: Market for Capital in Industry Y; Right Panel: Demand for Capital by a firm in industry Y

Employment of both factors increases in industry X (from L_0 to L_1 and from K_0 to K_1 , respectively). Simultaneously, the contraction of industry Y and exit of firms reduce the demand for both factors in industry Y. The demand curves D_{L_y} and D_{K_y} shift inward. The wage rate w_y and the rental rate r_y in industry Y fall. At this stage, disequilibrium

prevails in the factor markets because factor prices differ across industries: $w_x > w_y$ and $r_x > r_y$. This creates an incentive for factor mobility.

Step 4 — Factor Mobility and Restoring Equilibrium

With perfect factor mobility, the owners of labour and capital withdraw their services from industry Y (where w and r are lower) and offer them to firms in industry X (where w and r are higher). This causes: Supply of factors in industry X shifts rightward (downward), putting downward pressure on w_x and r_x ; Supply of factors in industry Y shifts leftward (upward), putting upward pressure on w_y and r_y . This process continues until factor prices are equalised across industries: $w_x = w_y = w^*$ and $r_x = r_y = r^*$. Due to the different factor intensities of the two industries

($\left(\frac{K}{L}\right)_x < \left(\frac{K}{L}\right)_y$), the new equilibrium factor prices differ from the initial ones: the demand of industry X for labour is stronger than for capital (X is more labour-intensive); the release of labour by Y is smaller than X requires; the release of capital by Y is larger than X needs. Therefore, in the new equilibrium: $w^* > w_0$ (wage is higher) and $r^* < r_0$ (rental is lower).

Step 5 — Final Product Market Equilibrium

Entry in industry X shifts the supply curve downward, reducing price from P_1 to P_2 (still above the original P_0 , since X is an increasing-cost industry). Exit from industry Y shifts supply curve upward, raising price. Firms in both industries earn normal profits in the long run. Individual firms adjust their outputs and factor demands in response to the new factor prices.

Question 13: *Explain the production equilibrium/consumption equilibrium/simultaneous equilibrium in a simple general equilibrium model through Edgeworth box diagram.*

In the two-factor (K, L), two-commodity (X, Y), two-consumer (A, B) general equilibrium model, three static properties characterise a general equilibrium state. These properties are called the marginal conditions of Pareto optimality. Each can be analysed through a specific Edgeworth box diagram.

(a) Production Equilibrium (Edgeworth Box of Production)

The Edgeworth box of production is constructed as follows. On the horizontal axis, measure the total quantity of labour L. On the vertical axis, measure the total quantity of capital K. The width of the box represents the total L; the height represents the total K. Any point inside the box represents a specific allocation of K and L between the two industries. Isoquants of commodity X are plotted with origin at the south-west corner. Higher X-isoquants lie further to the northeast. Isoquants of commodity Y are plotted with origin at the north-east corner, inverted. Higher Y-isoquants lie further to the southwest.

The locus of tangency points between X and Y isoquants is the Edgeworth contract curve of production. At each point on this curve, the slopes of the isoquants are equal:

$$MRTS_{L,K}^X = MRTS_{L,K}^Y$$

This equality is the condition for efficient allocation of factors between firms. Points off the contract curve are inefficient: a reallocation of factors can increase the output of at least one commodity without reducing the other.

In perfect competition, the factor prices w and r define the slope of isocost lines ($\frac{w}{r}$). Firms, being profit maximisers, choose the factor combination where the isoquant is tangent to the isocost line. Since all firms face the same factor prices, the production equilibrium occurs at the point on the contract curve where:

$$MRTS_{L,K}^X = MRTS_{L,K}^Y = \frac{w}{r}$$

This point defines the allocation of K and L between the two industries, and hence the specific quantities of X and Y produced.

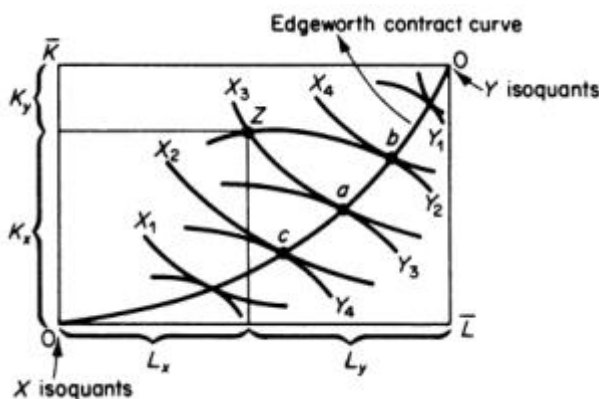


Figure – Edgeworth Box of Production with Contract Curve

The production possibility curve (PPC) is derived by mapping all points on the contract curve from factor space into output space. Each point on the PPC represents a Pareto-efficient combination of X and Y.

(b) Consumption Equilibrium (Edgeworth Box of Exchange)

For a given product-mix (Y_e, X_e) — determined by the production equilibrium — construct the Edgeworth box of exchange. The width of the box is X_e (the total quantity of X produced); the height is Y_e (the total quantity of Y). A's indifference curves are plotted from the south-west corner; B's from the north-east corner.

The locus of tangency points between A's and B's indifference curves is the Edgeworth contract curve of exchange. At each point on this curve:

$$MRS_{x,y}^A = MRS_{x,y}^B$$

This is the condition for efficient distribution of commodities between consumers. Points off this curve are inefficient: a redistribution of goods can increase at least one consumer's utility without reducing the other's.

With perfect competition, consumers face the same commodity prices P_x and P_y . The budget line has slope $-\frac{P_x}{P_y}$.

Consumer equilibrium occurs at the point on the contract curve where:

$$MRS_{x,y}^A = MRS_{x,y}^B = \frac{P_x}{P_y}$$

At this point (denoted T in the textbook), consumer A buys OM of X and ON of Y, reaching utility level A_2 . Consumer B buys the remaining MX_e of X and NY_e of Y, reaching utility level B_4 .

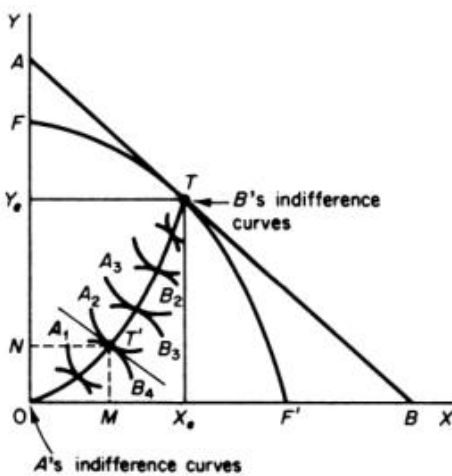


Figure – Edgeworth Box of Exchange for Product-Mix (Y_e , X_e)

(c) Simultaneous Equilibrium

General equilibrium of the entire system requires that the production plans of firms are consistent with the consumption plans of households. This is captured by the third marginal condition:

$$MRPT_{x,y} = MRS_{x,y}^A = MRS_{x,y}^B$$

The $MRPT$ (slope of the PPC) shows the rate at which Y can be transformed into X in production. The MRS (slope of the indifference curves) shows the rate at which consumers are willing to exchange Y for X. If these rates are unequal, there is a disequilibrium: either the economy produces too much of one good and too little of the other, relative to consumer preferences.

For example, if $MRPT_{x,y} = \frac{2Y}{X}$ and $MRS_{x,y} = \frac{1Y}{X}$, the economy can produce 2 units of Y by sacrificing 1 unit of X, while consumers are only willing to exchange 1 unit of Y for 1 unit of X. The economy produces too much X and too little Y. Firms must reduce X and increase Y to restore equilibrium.

At the simultaneous equilibrium point T on the PPC, the slope of the PPC equals the common MRS . In perfect competition, this is automatically satisfied because:

$$MRPT_{x,y} = \frac{MC_x}{MC_y} = \frac{P_x}{P_y} = MRS_{x,y}^A = MRS_{x,y}^B$$

The three conditions can be stated compactly:

- (1) $MRTS_{L,K}^x = MRTS_{L,K}^y = \frac{w}{r}$ (production equilibrium);
- (2) $MRS_{x,y}^A = MRS_{x,y}^B = \frac{P_x}{P_y}$ (consumption equilibrium);
- (3) $MRPT_{x,y} = \frac{P_x}{P_y}$ (simultaneous equilibrium, connecting (1) and (2)).

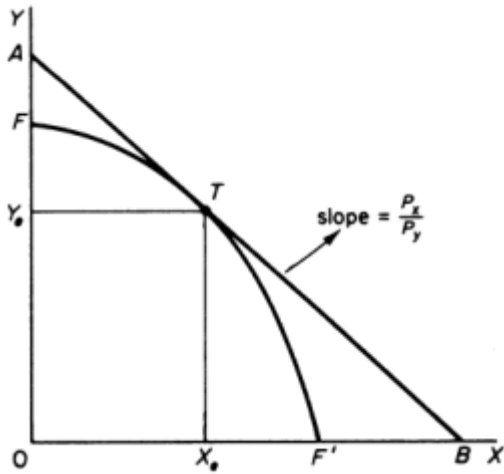


Figure – General Equilibrium of Production with Perfect Competition

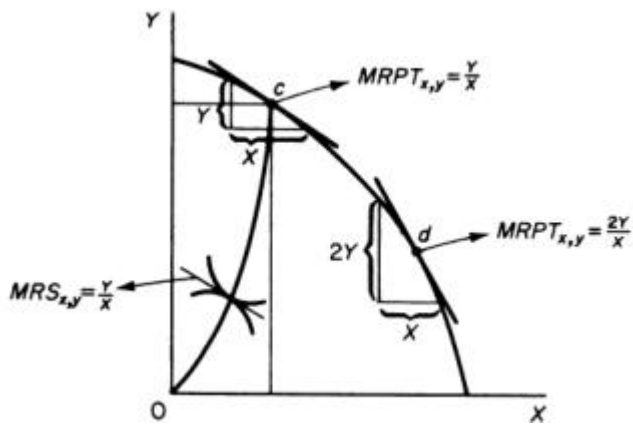


Figure – Disequilibrium When $MRPT \neq MRS$

Question 14: *Define general equilibrium. In what respect does it differ from partial equilibrium? Examine with example.*

Definition of General Equilibrium

A general equilibrium is defined as a state in which all markets and all decision-making units are in simultaneous equilibrium. More formally, a general equilibrium exists if each market in the economy is cleared at a positive price (i.e., there is neither excess demand nor excess supply), with each consumer maximising satisfaction (utility) and each firm maximising profit. The scope of general equilibrium analysis is the examination of how this state can be reached — that is, how prices are determined simultaneously in all markets so that the plans of all economic agents are mutually consistent.

General equilibrium emerges from the solution of a simultaneous equation model. In the Walrasian system, the unknowns are the prices of all factors and all commodities, and the quantities purchased and sold by each consumer and each producer. The equations are of two types: behavioural equations (demand and supply functions derived from optimisation) and clearing-the-market equations. The system captures the fundamental interdependence of all markets: consumers' demands depend on incomes, incomes depend on factor prices, factor prices depend on demand for inputs by firms, and firms' input demands depend on demand for final goods.

Definition of Partial Equilibrium

Partial equilibrium analysis, developed by Alfred Marshall, concentrates on decisions in a particular segment of the economy in isolation, under the ceteris paribus assumption. In this approach, the price and quantity in each market are determined by demand and supply curves drawn on the ceteris paribus clause. Each market is regarded independently of the others. The partial equilibrium approach is also known as the 'Marshallian approach'.

Key Differences

- 1. Scope:** Partial equilibrium examines one market at a time (e.g., the market for good X), holding all other market conditions constant. General equilibrium examines all markets simultaneously (e.g., markets for X, Y, labour, and capital together).
- 2. Interdependence:** Partial equilibrium ignores interdependence between markets. General equilibrium explicitly accounts for the circular interdependence: consumers' demands depend on incomes, incomes depend on factor prices, factor prices depend on demand for inputs by firms, and firms' demands depend on demand for final goods.
- 3. Price determination:** In partial equilibrium, price is determined by supply and demand in a single market. In general equilibrium, prices in all markets are simultaneously determined as a solution to a system of simultaneous equations.
- 4. Number of variables:** Partial equilibrium deals with a few variables (price and quantity in one market). General equilibrium involves millions of variables (prices and quantities of all commodities and factors, and their allocation among all consumers and producers).
- 5. Completeness of Results:** Partial equilibrium results are valid only for the market under study. General equilibrium results account for all cross-market effects.

Example

Partial equilibrium analysis:

A tax on commodity X shifts the supply curve upward. The equilibrium price rises (but by less than the full tax), the quantity falls, consumer surplus decreases, producer surplus decreases, and the government collects tax revenue. Deadweight loss is created. The analysis stops here.

General equilibrium analysis:

In addition to the direct effects on the X market, the tax triggers a cascade of adjustments across the economy:

- (a) The market for substitute good Y: Demand for Y increases (since X is now more expensive). P_Y rises and Q_Y increases.
- (b) Factor markets: As X production decreases and Y production increases, the demand for factors shifts. If X and Y have different factor intensities, factor prices change. If X is labour-intensive and Y is capital-intensive, the demand for labour decreases and the demand for capital increases, leading to a decrease in wages and an increase in the rental rate of capital.
- (c) Income distribution: The change in factor prices affects the distribution of income between labour and capital owners. Workers (whose wages have fallen) reduce their demand for both X and Y. Capital owners (whose rental income has risen) increase their demand. These income effects feed back into the product markets.
- (d) General equilibrium: A new set of equilibrium prices (P_X^*, P_Y^*, w^*, r^*) emerges, at which all four markets clear simultaneously. The final effects on prices and quantities differ from the partial equilibrium predictions because of the cross-market feedback effects.

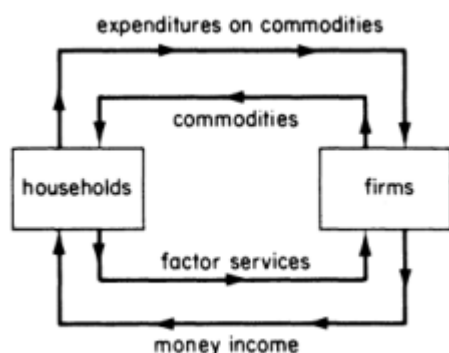


Figure – Circular Flows in a Two-Sector Economy

Question 15: *Present a $2 \times 2 \times 2$ Walrasian system. Does the general equilibrium solution exist in such a model? Explain.*

The $2 \times 2 \times 2$ Walrasian System

The Walrasian system, developed by the French economist Léon Walras (1834–1910), is the most ambitious general equilibrium model. In the simplified $2 \times 2 \times 2$ version, the economy consists of: two consumers (A and B) who buy commodities and supply factor services; two commodities (X and Y), each produced by one firm; two factors of production (K and L) owned by the consumers.

Unknowns (18 total):

Quantities demanded of X and Y by consumers: $2 \times 2 = 4$

Quantities supplied of K and L by consumers: $2 \times 2 = 4$

Quantities demanded of K and L by firms: $2 \times 2 = 4$

Quantities of X and Y supplied by firms: 2

Prices of commodities (P_x, P_y): 2

Prices of factors (w, r): 2

Equations (18 total):

Consumer demand functions: $2 \times 2 = 4$

Consumer factor supply functions: $2 \times 2 = 4$

Firm factor demand functions: $2 \times 2 = 4$

Firm commodity supply functions: 2

Commodity market clearing: 2

Factor market clearing: 2

Does a Solution Exist?

Although the number of equations equals the number of unknowns (18 each), a general equilibrium solution is not guaranteed for several reasons:

- 1. Redundant equation:** One equation is not independent of the others (proved via Walras's Law: if aggregate expenditures equal aggregate income as an identity, then if $N-1$ markets clear, the N th must also clear). This means the number of independent equations is 17, less than the 18 unknowns. Prices can only be determined as ratios.
- 2. Numeraire device:** By choosing one commodity as numeraire (setting its price = 1), we reduce unknowns to 17, matching the 17 independent equations. However, this determines only relative prices, not absolute price levels.
- 3. Existence proof:** Arrow and Debreu (1954) proved that a general equilibrium exists in perfectly competitive markets with no indivisibilities and no increasing returns to scale, provided production and demand functions are continuous and well-behaved (convex). The proof relies on fixed-point theorems rather than simple equation counting.
- 4. Counting is insufficient:** Equality of equations and unknowns is neither necessary nor sufficient for existence. The system may have no solution, a unique solution, or multiple solutions depending on the shapes and properties of the underlying functions.

Under the assumptions of the model (perfect competition, convex preferences, constant returns to scale, no externalities), a general equilibrium solution does exist in the $2 \times 2 \times 2$ model, as proved by Arrow and Debreu. However, this existence depends critically on the restrictive assumptions. The system determines relative prices and quantities uniquely; absolute prices require the introduction of money.

Question 16: *What are the assumptions of a $2 \times 2 \times 2$ model? How does general equilibrium is reached in such a model?*

Assumptions

1. Two factors of production (L and K) in given (fixed) quantities; homogeneous and perfectly divisible.
2. Two commodities (X and Y) produced by two firms; technology given with smooth, convex-to-origin isoquants and constant returns to scale. Production functions are independent (no externalities).
3. Two consumers (A and B) with ordinal indifference curves convex to the origin, exhibiting diminishing MRS. Consumer choices are independent (no bandwagon, snob, or Veblen effects).
4. Consumers maximize utility subject to income constraint; firms maximize profit subject to production function constraint.
5. Factors are owned by consumers.
6. Full employment of factors; all incomes are spent.
7. Perfect competition in commodity and factor markets; all agents face the same prices (P_x , P_y , w , r).

How General Equilibrium Is Reached

The process begins from an initial equilibrium state that is disturbed by an exogenous change (e.g., a shift in consumer tastes).

1. Product market adjustment

Suppose demand for X increases. P_x rises, creating excess profits. Demand for substitute Y decreases; P_y falls, creating losses. Firms in X expand output; firms in Y contract. The economy moves along the PPC from the initial to a new product-mix.

2. Factor market adjustment

Expansion of X increases demand for L and K in industry X (D_{L_x} and D_{K_x} shift right; w_x and r_x rise). Contraction of Y reduces demand for L and K in industry Y (D_{L_y} and D_{K_y} shift left; w_y and r_y fall).

3. Factor mobility

Factor owners withdraw services from Y and seek employment in X where factor prices are higher. Supply of factors in X shifts rightward (reducing w_x and r_x); supply in Y shifts leftward (increasing w_y and r_y). This process continues until factor prices are equalized: $w_x = w_y = w^*$; $r_x = r_y = r^*$.

4. Final equilibrium

Due to different factor intensities ($\left(\frac{K}{L}\right)_x < \left(\frac{K}{L}\right)_y$), the new equilibrium wage w^* is higher than initially, and r^* is lower. Entry in X shifts S_x down (P_x falls but stays above P_0); exit from Y shifts S_y up (P_y rises but stays below original). Firms in both industries earn normal profits. All four markets clear simultaneously.

The key mechanism is factor mobility combined with entry and exit. The system converges because perfect competition ensures that market signals (price changes) are transmitted correctly and agents respond optimally.

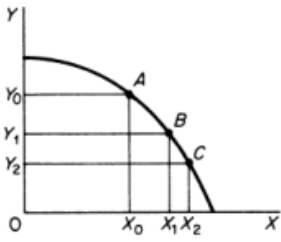


Figure – Movement Along the Production Possibility Curve

Question 17: Show that $MRPT_{x,y} = MC_x/MC_y$

Derivation

The Marginal Rate of Product Transformation (MRPT) is defined as the negative of the slope of the production possibility curve (PPC):

$$MRPT_{x,y} = -\frac{dY}{dX}$$

By definition, marginal cost is the change in total cost per unit change in output:

$$MC_x = \frac{d(TC_x)}{dX}$$

$$MC_y = \frac{d(TC_y)}{dY}$$

Dividing these expressions:

$$\frac{MC_x}{MC_y} = \left[\frac{d(TC_x)}{dX} \right] / \left[\frac{d(TC_y)}{dY} \right] \dots (1)$$

Now, the change in total cost for each commodity equals the sum of changes in factor payments:

$$d(TC_x) = w \cdot dL_x + r \cdot dK_x \dots (2)$$

$$d(TC_y) = w \cdot dL_y + r \cdot dK_y \dots (3)$$

To remain on the PPC (i.e., to maintain full employment of factors), the factors released from decreasing Y must be absorbed by increasing X:

$$dL_x = -dL_y \dots (4)$$

$$dK_x = -dK_y \dots (5)$$

Substituting (4) into (2):

$$d(TC_x) = w \cdot (-dL_y) + r \cdot (-dK_y) = -[w \cdot dL_y + r \cdot dK_y] \dots (6)$$

Now substituting (3) and (6) into (1):

$$\frac{MC_x}{MC_y} = -[w \cdot dL_y + r \cdot dK_y] / [w \cdot dL_y + r \cdot dK_y]$$

$$\frac{MC_x}{MC_y} = -1 \left[\frac{dY}{dX} \right] = -\frac{dY}{dX} \dots (7)$$

But from the definition:

$$MRPT_{x,y} = -\frac{dY}{dX}$$

Therefore:

$$\frac{MC_x}{MC_y} = MRPT_{x,y}$$

$$MRPT_{x,y} = \frac{MC_x}{MC_y} \quad \text{Q.E.D.}$$

Intuition

The MRPT shows the rate at which the economy can transform Y into X. To produce one more unit of X, the economy must absorb additional factors whose cost equals MC_x . Simultaneously, to release these factors, the economy must reduce Y by an amount whose cost saving equals MC_y . The ratio MC_x/MC_y precisely captures the rate at which Y must be sacrificed (in cost terms) to obtain an additional unit of X.

Corollary (in Perfect Competition)

Since firms set $P = MC$ in perfect competition:

$$MRPT_{x,y} = \frac{P_x}{P_y}$$

This connects the production side (MRPT) with the consumption side ($MRS = \frac{P_x}{P_y}$), ensuring that the third Pareto optimality condition ($MRPT = MRS$) is satisfied under perfect competition.

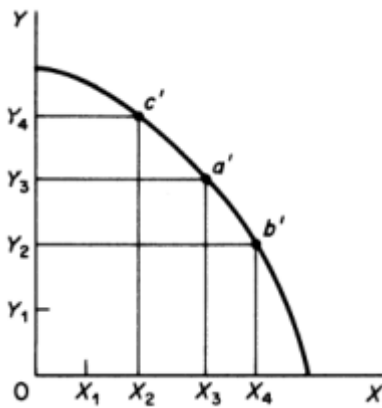


Figure – Production Possibility Curve Derived from Edgeworth Contract Curve